# Craig Fouts

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## CORE SKILLS

Python | Julia | C/C++ | PyTorch | Scikit-Learn | Scipy | Pandas | AnnData | Muon | Sphinx | SDL3 | Probabilistic Modelling | Statistical Inference | Interpretable Machine Learning | System Identification | Predictive Control | Functional Genomics | Spatial Transcriptomics

## EDUCATION

### Columbia University | MSc

**Applied Mathematics** **2022 – 2023**

Completed a graduate research assistantship in the New York Genome Center Technology Innovation Laboratory.

### The Ohio State University | BSc

**Computer Science/Mathematics** **2018 – 2022**

Received honors research distinction for original research and publication in collaboration with Google Research.

## EXPERIENCE

### Uppsala University | Vicković Group, Science for Life Laboratory

**Computational Research Engineer** **Oct 2024 – Sep 2025**

Designed and developed an original nonparametric topic model called **ATLAS** that elucidates anatomical niches and pathological motifs in spatial transcriptomics datasets based on gene expression profile and spatial distribution. The model adaptively learns the number of spatially-resolved niches while identifying shared motifs across histological batches.

### New York Genome Center | Technology Innovation Laboratory

**Associate Computational Biologist II** **Jan 2024 – Sep 2024**

Developed a novel probabilistic generative model called **sceLDA** that clusters anatomical niches in spatial proteomics datasets based on cell type composition and spatial distribution. The model is part of **Hudson**, a spatial-omics ecosystem for repurposing decommissioned Illumina HiSeq 2500 sequencers as accessible immunofluorescent imaging platforms.

### The Ohio State University | Translational Data Analytics Institute

**Undergraduate Research Assistant** **Aug 2021 – Sep 2022**

Developed a computational pipeline for aggregating, annotating, and analyzing multi-modal (CO2, PM2.5/10, noise, etc.) data collected via environmental sensors used to study the effects of aircraft combustion engines in urban neighborhoods. The pipeline uses dynamic time warping to align several data streams based on location and ambient conditions.

## PUBLICATIONS

### Growing Steerable Neural Cellular Automata

Ettore Randazzo, Alexander Mordvintsev, **Craig Fouts** (24 July, 2023). Growing Steerable Neural Cellular Automata. Proceedings of ALIFE 2023: The 2023 Conference on Artificial Life (pp. 2). ASME. https://doi.org/10.1162/isal_a_00564

### Growing Isotropic Neural Cellular Automata

Alexander Mordvintsev, Ettore Randazzo, **Craig Fouts** (18 July, 2022). Growing Isotropic Neural Cellular Automata. Proceedings of ALIFE 2022: The 2022 Conference on Artificial Life (pp. 65). ASME. https://doi.org/10.1162/isal_a_00552

## ACCOLADES

### Honors

**The Ohio State University:** Magna Cum Laude | Honors Research Distinction **2022**
**Granville High School:** Cum Laude Society | National Honor Society | Sociedad Honoraria Hispánica **2017**
**Scouting America:** Eagle Scout **2016**

### Competitions

**HackOHI/O Hackathon:** 1st Place Grand Prize | Microsoft Challenge Winner | People's Choice Award **2021**
**Ohio State FEH Honors Robotics Competition:** 2nd Place Outstanding Achievement in Innovation **2019**
**OMEA Solo & Ensemble:** Rank 1 Class A Violin Solo Performance **2016** & **2017**

## COURSEWORK

### Columbia University

Applied Statistics III (A) | Machine Learning for Functional Genomics (A) | Advanced Linear Algebra (A+) **2023**
Numerical Algebra & Optimization (A) | Partial Differential Equations (A-) **2022**

### The Ohio State University

Discrete Mathematical Models (A) | Quantitative Neuroscience (A) | Computer Networking (A) **2022**
Mathematical Statistics II (A) | Advanced Artificial Intelligence (A) | Programming Languages (A) **2021**
Data Structures & Algorithms (A) | Experimental Physics (A) | Intermediate Mechanics (A-) **2020**
Ordinary Differential Equations (A) | Honors Physics Electricity & Magnetism (A) **2019**
Honors Real Analysis (A) | Honors Psychology (A) **2018**